CURIOUS SPAROOWS · ENCYCLOPEDIC INDEX

GARDENING

How a seed in the soil became the first step of every civilisation

Kinds of gardens · garden traditions of India and the world · the plant world · soil, water, light and climate · techniques · plant health · plant cards · history · people · shows and awards · ideas · practical guide

Parts	Sections	Sub-sections	Entries	Register rows
16	42	35	307	118

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How to Use This Booklet

This booklet is a map of gardening: kinds of gardens, garden traditions of India and the world, the plant world, soil, water, light and climate, techniques, plant health, plant cards, history, people, shows and awards, ideas and a practical guide. Every entry tells you, in one or two lines, what the thing is and why it matters.

Every entry has an ID (such as GRD-13.7 for grafting). The same ID is the entry's address on curioussparoows.com, where each one is growing into its own page.

How this booklet is organised

Parts: the intro questions, the master map, kinds of gardens, traditions, the plant world, growing conditions, techniques, plant health, registers, history, people and awards, ideas, the practical guide, reviews, navigation and sources.

Sections (GRD-01 to GRD-31): the main topics, each opening with a *Before you start* box.

Sub-sections and entries (☐): one or two lines each; tick the box when you have grown or visited.

Registers: 30 plant cards, the timeline, shows and awards, and a monthly calendar. They are a sourced selection; the full lists grow on the website.

How to explore it

Step 1. Read Part 0, six two-minute questions.

Step 2. Find the kind of garden that fits your space (Part 2).

Step 3. Pick three plants from the plant cards (Part 8).

Step 4. Follow the 30-day first garden (Part 12).

Step 5. Tick ☐ as you grow, and keep a garden diary.

Our promise

Knowledge belongs to the people who created it. Every fact here stands on a source listed in Part 15, and every "latest" fact says when it was checked. Where something could not be verified, we say so instead of guessing. Found a mistake? Tell us, and we will fix it and thank you.

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PART 0 — THE INTRODUCTION TWIG: HOW DID A SEED IN THE SOIL BECOME THE FIRST STEP OF EVERY CIVILISATION?

About this Part. This is the doorway to the Gardening section of the website, in the tree-twig format of the Curious Sparoows library: one Gateway page, six Leaves and one Recap. Each Leaf answers one question in about two minutes (250 to 400 words), gives its short answer first and ends with a hook to the next question.

GRD-00 The Gateway: How did a seed in the soil become the first step of every civilisation?

Web address: /gardening/seed-to-civilisation/ · **Keywords:** gardening for beginners, history of gardening, types of gardening, kitchen garden India, why gardening is good for you

Every city on Earth stands on a quiet invention: someone, about 11,000 years ago, saved a handful of seeds, put them in the ground and waited. Farming gave us villages, villages gave us cities, and cities gave us writing, mathematics and gardens made just for beauty. When you sow a coriander seed in a pot on your balcony, you are repeating the first step of civilisation.

This short course asks one question in six steps: **what happens between a seed and a garden, and why does it change us?** It goes from the first farmers of the Fertile Crescent and Mehrgarh to the four-part gardens of the Mughals, from the living world inside a spoonful of soil to the science of why gardeners feel calmer. Each page takes about two minutes.

By the end you will be able to:

- Say where and when farming began, including in South Asia.
- Explain how a seed knows which way to grow.
- Describe why healthy soil is alive, and how compost feeds it.
- Read the plan of a Mughal *char bagh* and say what it means.
- Plan a small food garden for a balcony.
- Explain, with evidence, why gardening is good for body and mind.

GRD-00.01 Who planted the first garden?

> **Short answer:** Nobody knows the first gardener's name, but farming began independently in several places: the Fertile Crescent of West Asia about 11,500 years ago, China, New Guinea, Mesoamerica, the Andes, Africa and South Asia. At Mehrgarh in today's Balochistan, people grew barley and wheat and herded cattle by about 7000 BCE.

Keywords: first farmers, Neolithic revolution, Mehrgarh, origin of agriculture, first garden in history

The Neolithic revolution

The archaeologist V. Gordon Childe called it the "Neolithic revolution" (1936): the change from gathering wild food to growing it. It was not one sudden event but a slow partnership. Wild wheat drops its seeds when ripe; a few mutant plants kept them on the stalk. People harvested those, sowed them again and, without meaning to, bred crops that need humans to plant them.

Many cradles

- **West Asia**: emmer and einkorn wheat, barley, lentils, peas, flax (about 9500 BCE).
- **China**: rice in the Yangtze valley, millet in the Yellow River valley.
- **New Guinea**: taro and bananas, about 7000 years ago.

- **The Americas**: maize, beans and squash (the "three sisters"), potatoes in the Andes.
- **Africa**: sorghum, pearl millet, African rice, yams.
- **South Asia**: Mehrgarh's barley and wheat; later rice, millets, mung bean, cotton and sesame in the Indus and Ganga lands.

Gardens for beauty

Once food was secure, people began to grow plants simply to enjoy them. Egyptian tomb paintings show walled gardens with pools (about 1400 BCE), and Mesopotamian kings boasted of their parks.

Remember this: Farming began in many places at once, and South Asia was one of them.

Next: A seed has no eyes. How does it know which way is up?

GRD-00.02 How does a seed know which way is up?

Short answer: Plants sense gravity with tiny starch grains called statoliths that settle at the bottom of special cells in the root tip. The plant hormone auxin then gathers on one side, so the root curves down and the shoot curves up. This is called gravitropism; growing towards light is phototropism.

Keywords: how seeds germinate, gravitropism, phototropism, auxin, seed germination for kids

Waking up

A seed is a baby plant with a packed lunch, wrapped in a coat. It waits (dormancy) until water, warmth and air tell it to wake. It drinks water, swells, splits its coat and sends out a root (the radicle) first, then a shoot.

Feeling gravity

In the root cap, cells called statocytes hold heavy starch grains. Whichever way you turn the seed, the grains sink to the "floor" of each cell, and the plant reads that as down. Charles Darwin and his son Francis showed in *The Power of Movement in Plants* (1880) that the tip of a seedling senses light and sends a signal down; that signal was later identified as auxin.

Try it

Soak a few mung beans, place them in a clear glass between wet tissue and the glass wall, and turn the glass on its side after three days. Within a day the roots bend downwards again.

Remember this: Roots go down and shoots go up because the plant feels gravity with starch grains and steers with a hormone.

Next: Roots grow down into soil. But what is soil, and why do gardeners call it alive?

GRD-00.03 Why is soil alive?

Short answer: Healthy soil is a living city. A single teaspoon can hold billions of microorganisms: bacteria, fungi, protozoa and tiny animals that break down dead matter, make nutrients available and build the crumbly structure that holds water and air.

Keywords: what is soil, soil health, living soil, soil microorganisms, compost and soil

What soil is made of

Roughly, a good garden soil is about half solid (minerals from weathered rock plus a few per cent of organic matter) and half pore space, shared by water and air. The minerals come in three sizes: sand (gritty), silt (silky) and clay (sticky). The mix of the three is the soil's texture (GRD-10).

The living part

- **Bacteria and fungi** decompose leaves and roots. Some bacteria (*Rhizobium*) live in the root nodules of peas and beans and fix nitrogen from the air.
- **Mycorrhizal fungi** join plant roots and stretch the root system many times over, trading water and phosphorus for sugar.

- **Earthworms**, which Darwin praised in his last book (1881), mix and aerate the soil.

Why it matters

The FAO says about 95 per cent of our food comes directly or indirectly from soil, and that soil forms very slowly: a centimetre can take hundreds of years. That is why compost, mulch and not leaving soil bare are the gardener's first duties.

Remember this: Feed the soil, and the soil feeds the plant.

Next: Gardens are not only for food. Why did the Mughals divide theirs into four?

GRD-00.04 Why do Mughal gardens have four parts?

Short answer: The Mughal *char bagh* ("four gardens") follows a Persian plan: two water channels cross at the centre and divide the garden into four. It echoes the four rivers of paradise described in the Qur'an (of water, milk, wine and honey) and in older Persian ideas; the word "paradise" itself comes from the Old Persian *pairidaēza*, a walled enclosure.

Keywords: char bagh, Mughal gardens, Persian garden, Shalimar Bagh, paradise garden meaning

From Persia to India

The first Mughal emperor, Babur, missed the gardens of Central Asia and laid out the Ram Bagh (Aram Bagh) at Agra in the 1520s. His successors built the great ones: Humayun's Tomb garden in Delhi (1560s), the Shalimar Bagh and Nishat Bagh in Srinagar (1600s), and the garden of the Taj Mahal, where the tomb stands at the end rather than the centre.

How to read a char bagh

- **Walls**: shelter from dust and the world outside.
- **Water**: channels, chutes (*chadar*) and fountains for coolness and sound; terraces on hillsides let water fall from level to level.
- **Four quarters**: fruit trees, flowers and shade.
- **The centre**: a pool, pavilion or tomb.

India's own garden ideas

India had gardens long before the Mughals: sacred groves (GRD-05.2), the *vatika* of Sanskrit poetry, temple flower gardens, and the homegardens of Kerala. The Mughal style joined this story; it did not begin it.

Remember this: A char bagh is a picture of paradise drawn with water.

Next: Few of us have palace gardens. Can a balcony feed a family?

GRD-00.05 Can you grow food on a balcony?

Short answer: Yes. A sunny balcony of a few square metres can give you herbs every day, a steady supply of greens, and seasonal tomatoes, chillies and beans, if it gets at least four to six hours of direct sun. It will not replace the market, but it can replace a lot of the herbs and greens you buy.

Keywords: balcony garden India, terrace garden, container vegetables, grow vegetables in pots, urban gardening

What decides success

- **Sun**: leafy greens and herbs cope with 3 to 4 hours; fruiting plants (tomato, chilli, brinjal) want 6 or more.
- **Containers**: bigger is easier. Greens need about 15 cm of soil depth; tomatoes and brinjal about 30 to 45 cm.
- **Potting mix**: never plain garden soil, which packs hard in pots. A common mix is one part soil, one part compost and one part coco peat or similar.
- **Water**: pots dry fast; water deeply when the top 2 cm is dry, and mulch.
- **Weight**: check how much load the balcony can carry before using many large pots.

Easy first crops

Coriander, methi (fenugreek), spinach, mint, curry leaf, tulsi, spring onion, green chilli, cherry tomato, and microgreens on a windowsill (ready in 7 to 14 days).

Remember this: Start with sun, a big pot and greens; harvest within a month.

Next: Gardeners often say gardening makes them calmer. Is there evidence?

GRD-00.06 Why does gardening make people calmer?

Short answer: Studies suggest it does. A 2017 meta-analysis of 22 studies (Soga, Gaston and Yamaura, *Preventive Medicine Reports*) found gardening linked with lower depression and anxiety and better life satisfaction. Explanations include gentle exercise, time outdoors in daylight, a sense of achievement, social contact and what psychologists call "soft fascination".

Keywords: gardening and mental health, horticultural therapy, benefits of gardening, gardening and stress, attention restoration

What the research says

- **Stress**: in a small Dutch study (Van den Berg and Custers, 2011), gardening after a stressful task lowered cortisol more than reading indoors.
- **Attention**: Rachel and Stephen Kaplan's *attention restoration theory* (1989) proposes that natural settings rest the tired, focused mind.
- **Views**: Roger Ulrich (*Science*, 1984) found that surgical patients with a window view of trees went home a little sooner and needed fewer strong painkillers than those facing a brick wall.
- **Community gardens** build friendships and food security.

Horticultural therapy

Horticultural therapy is gardening guided by a trained therapist for health goals: in hospitals, prisons, schools and care homes. The American Horticultural Therapy Association (1973) and the UK charity Thrive (1978) train practitioners.

A note of honesty

Many studies are small, and people who already feel well may garden more. The evidence is encouraging, not final.

Remember this: Gardening combines movement, nature, purpose and people, and each of these is good for the mind.

Next: The Recap ties the six questions together.

GRD-00.07 Recap: from one seed to a garden of the world

- **GRD-00.01**: Farming began in many places around 9500 BCE onward, including Mehrgarh in South Asia.
- **GRD-00.02**: Seeds sense gravity with starch grains and steer with auxin.
- **GRD-00.03**: Soil is alive; compost and mulch feed it.
- **GRD-00.04**: The char bagh is a Persian picture of paradise, built in India by the Mughals.
- **GRD-00.05**: A sunny balcony can keep a family in herbs and greens.
- **GRD-00.06**: Gardening is linked with lower stress and better mood.

Where to go next: the Master Map (Part 1), then any kind of garden that suits your space (Part 2). **Quick quiz:** What is a statolith? What does *pairidāēza* mean? Name two nitrogen-fixing crops.

PART 1 — THE MASTER MAP OF GARDENING

About this Part. The whole hierarchy on one page, so every kind of garden, plant and technique has an address. The website follows the same tree.

GRD-01 The hierarchy: from Forest to Leaf

Before you start — GRD-01. The Forest is "Gardening". It splits into seven Trees: kinds of gardens (what you grow and where), garden traditions (how cultures have shaped gardens), the plant world, the growing conditions (soil, water, light, climate), techniques, plant health, and the human side (history, people, ideas, practice). **See also:** GRD-00 (the Gateway).

☐ **GRD-01.1 Kinds of gardens**: food, ornamental, special-purpose, by space, and by method (Part 2).

☐ **GRD-01.2 Garden traditions**: India and the world (Part 3).

☐ **GRD-01.3 The plant world**: families, life cycles, plant types, native plants and seasons (Part 4).

☐ **GRD-01.4 Growing conditions**: soil, compost, water, light and climate (Part 5).

☐ **GRD-01.5 Techniques**: sowing, propagation, pruning, companion planting, seed saving (Part 6).

☐ **GRD-01.6 Plant health**: pests, diseases, deficiencies and integrated pest management (Part 7).

☐ **GRD-01.7 Registers**: plant cards, the timeline, shows and awards (Part 8).

☐ **GRD-01.8 The human side**: history (Part 9), people and awards (Part 10), ideas (Part 11), the practical guide (Part 12) and reviews (Part 13).

PART 2 — KINDS OF GARDENS

About this Part. Every kind of garden in the brief, sorted by what it grows, where it grows and how. A garden often belongs to several kinds at once (an organic balcony herb garden), and the website tags each one.

GRD-02 Food gardens

Before you start — GRD-02. Food gardening is the oldest kind. In India it is the *kitchen garden* or *poshan vatika* (nutrition garden), promoted by the government's POSHAN Abhiyaan and by Krishi Vigyan Kendras. **See also:** COK (Cooking), GRD-09 (seasons).

- ☐ **GRD-02.1 Kitchen and vegetable garden**: greens, roots, gourds and fruiting vegetables grown near the home; the classic Indian backyard *bari* and the English allotment.
- ☐ **GRD-02.2 Nutrition garden (poshan vatika)**: a planned plot with greens, a pulse, a fruit tree such as moringa or papaya, and a root crop, so a family gets vitamins A and C and iron through the year.
- ☐ **GRD-02.3 Fruit garden and orchard**: trees such as mango, guava, citrus, pomegranate and amla; temperate apples, pears and stone fruit in the hills; dwarf and grafted trees for small spaces.
- ☐ **GRD-02.4 Herb garden**: culinary herbs (coriander, mint, basil, curry leaf, ajwain, lemongrass, rosemary, thyme).
- ☐ **GRD-02.5 Medicinal garden**: plants of Ayurveda, Unani, Siddha and folk medicine (tulsi, ashwagandha, giloy, aloe, brahmi, neem); the National Medicinal Plants Board promotes home herbal gardens. *Caution:* growing a plant is not the same as safe dosing; ask a qualified practitioner.
- ☐ **GRD-02.6 Spice garden**: chilli, turmeric, ginger, black pepper on a support tree, cardamom in shade (Kerala and Karnataka hills).
- ☐ **GRD-02.7 Microgreens and sprouts**: seedlings harvested at 7 to 14 days; the fastest food garden on a windowsill.
- ☐ **GRD-02.8 Mushroom growing**: oyster and button mushrooms on straw or compost in a dark room; strictly speaking fungi, not plants.

GRD-03 Ornamental and special-purpose gardens

Before you start — GRD-03. Ornamental gardens are grown for beauty, scent, shade or meaning. Special-purpose gardens serve wildlife, water, health or learning.

☐ GRD-03.1 Beauty and design

- ☐ **GRD-03.1a Flower garden**: annual beds (marigold, zinnia, petunia), perennial borders, cutting gardens; India's winter flower season in the plains runs October to March.
- ☐ **GRD-03.1b Rose garden (rosarium)**: the Zakir Hussain Rose Garden in Chandigarh (1967) is among Asia's largest.
- ☐ **GRD-03.1c Ornamental foliage and shrub garden**: crotons, coleus, hibiscus, bougainvillea, ixora.
- ☐ **GRD-03.1d Lawn and turf**: grass for play and calm; in India doob grass (*Cynodon dactylon*) and its improved forms; a thirsty choice in dry regions.
- ☐ **GRD-03.1e Bonsai**: miniature trees trained in shallow pots by pruning and wiring; Chinese *penjing* came first, Japan refined it into bonsai.
- ☐ **GRD-03.1f Succulents and cacti**: water-storing plants; easy in sunny, dry spots and the first plants of many indoor gardeners.
- ☐ **GRD-03.1g Topiary and hedges**: shrubs clipped into shapes; Roman in origin, revived in Renaissance and English gardens.
- ☐ **GRD-03.1h Rock garden**: alpine and dry plants among stones; also the name of Nek Chand's sculpture garden in Chandigarh (GRD-20).

☐ GRD-03.2 Water, glass and moss

☐ **GRD-03.2a Aquatic and water garden**: ponds and tubs with lotus, water lily, water hyacinth (invasive in India: never release into streams), and fish that eat mosquito larvae.

☐ **GRD-03.2b Terrarium**: a garden in a glass jar; the closed Wardian case (1830s) carried tea and rubber plants across oceans (GRD-19).

☐ **GRD-03.2c Kokedama**: a Japanese "moss ball" with a plant's roots wrapped in soil and moss, hung on a string.

☐ **GRD-03.2d Moss and fern garden**: shade gardens in humid places.

☐ GRD-03.3 Gardens with a purpose

☐ **GRD-03.3a Wildlife and pollinator garden**: native flowers, water and shelter for bees, butterflies and birds; host plants such as curry leaf and citrus for swallowtail caterpillars.

☐ **GRD-03.3b Rain garden**: a shallow planted dip that catches roof and road run-off so it soaks in instead of flooding drains.

☐ **GRD-03.3c Xeriscape and desert garden**: water-wise design with native and drought-tolerant plants; natural in Rajasthan (khejri, rohida, ber, kair, aak).

☐ **GRD-03.3d Therapeutic and sensory garden**: gardens in hospitals and care homes designed for healing; scent, touch and wheelchair-height beds (GRD-22).

☐ **GRD-03.3e School garden**: learning by growing; India's school nutrition gardens under PM POSHAN.

☐ **GRD-03.3f Memorial and sacred garden**: temple gardens, *nakshatra vanas*, memorial groves and cemetery gardens (GRD-05).

☐ **GRD-03.3g Botanical garden and arboretum**: living collections for science and conservation (GRD-19.4).

GRD-04 Gardens by space and by method

Before you start — GRD-04. The same tomato can be grown in a field, a pot, a pipe of water or a wall pocket. This section sorts gardens by where they are and by the system used.

☐ GRD-04.1 By space

☐ **GRD-04.1a Backyard and front garden**: the classic home garden.

☐ **GRD-04.1b Balcony and terrace garden**: containers, grow bags and raised beds on roofs; check load-bearing and waterproofing first.

☐ **GRD-04.1c Container and pot garden**: clay, plastic, fabric grow bags, recycled tins; drainage holes are non-negotiable.

☐ **GRD-04.1d Indoor plants**: money plant (pothos), snake plant, ZZ plant, peace lily, spider plant; needs bright indirect light. The famous NASA Clean Air Study (Wolverton, 1989) was done in sealed chambers; later reviews (Cummings and Waring, 2019) found that houseplants do little to clean air in real homes compared with ventilation. Grow them for joy, not as air purifiers.

☐ **GRD-04.1e Vertical garden and green wall**: pockets, panels or trellises on walls; Patrick Blanc's *mur végétal* made it famous.

☐ **GRD-04.1f Rooftop and urban farming**: city rooftops, Mumbai's and Bengaluru's terrace farming communities; Singapore and New York rooftop farms.

☐ **GRD-04.1g Community garden and allotment**: shared plots; Britain's allotments, Cuba's *organopónicos*, and city community gardens in Indian metros.

☐ GRD-04.2 By method

☐ **GRD-04.2a Organic gardening**: no synthetic pesticides or fertilisers; India's standard is the NPOP, and Sikkim was declared a fully organic state in 2016.

☐ **GRD-04.2b Natural farming**: low-input methods such as Masanobu Fukuoka's "do-nothing" farming (*The One-Straw Revolution*, 1975) and Subhash Palekar's zero-budget natural farming with *jeevamrut*; India's National Mission on Natural Farming (2024). Yield claims are debated; see GRD-E.

☐ **GRD-04.2c Permaculture**: "permanent agriculture" designed on natural ecosystems, coined by Bill Mollison and David Holmgren (1978); guilds, zones and food forests.

☐ **GRD-04.2d Hydroponics**: growing in nutrient-rich water instead of soil (NFT, deep water culture, wick systems); popular for lettuce and herbs.

☐ **GRD-04.2e Aquaponics**: fish and plants in one loop: fish waste feeds plants and plants clean water for the fish; the Aztec *chinampas* are an ancient cousin.

☐ **GRD-04.2f Square-foot and raised-bed gardening**: Mel Bartholomew's grid method (1981) for small, intensive beds.

☐ **GRD-04.2g No-dig gardening**: mulch and compost on top instead of digging; championed by Charles Dowding (UK) and Ruth Stout's deep-mulch method (US).

☐ **GRD-04.2h Biodynamic gardening**: Rudolf Steiner's 1924 lectures, with lunar calendars and preparations; popular with some growers but its special preparations lack scientific support.

☐ **GRD-04.2i Polyhouse and protected cultivation**: greenhouses, shade nets and polytunnels; widely subsidised by India's horticulture missions.

PART 3 — GARDEN TRADITIONS OF INDIA AND THE WORLD

About this Part. How cultures have shaped gardens, starting with India. Each tradition gets its idea, its look and its famous examples. Most examples can be visited; check timings locally.

GRD-05 Garden traditions of India

Before you start — GRD-05. India's gardens are older and more varied than the Mughal gardens alone: there are sacred groves, temple flower gardens, palace gardens, homegardens and colonial botanical gardens. Sanskrit texts such as the *Vrikshayurveda* ("science of plant life") of Surapala (about 10th century CE) give advice on sowing, watering and plant care. **See also:** GRD-00.04.

- ☐ **GRD-05.1 Vrikshayurveda**: Sanskrit treatises on tree care, including Surapala's text and chapters in Varahamihira's *Brihat Samhita* (6th century CE); they describe *kunapajala*, a fermented liquid manure still tested by organic farmers today.
- ☐ **GRD-05.2 Sacred groves**: patches of forest protected by local deities and custom: *devrai* in Maharashtra, *kavu* in Kerala, *orans* in Rajasthan, *sarna* in Jharkhand, *law kyntang* in Meghalaya (Mawphlang). India has many thousands; they are biodiversity refuges.
- ☐ **GRD-05.3 Nakshatra and Navagraha vanas**: gardens planting the tree linked with each of the 27 lunar mansions or the nine planets (for example amla, peepal, khair); a living tradition promoted by forest departments and temples.
- ☐ **GRD-05.4 Temple flower gardens (nandavanam)**: gardens that grow jasmine, lotus, hibiscus, tulsi and bel for daily worship, especially in South India.
- ☐ **GRD-05.5 Buddhist and Jain gardens**: the Bodhi tree at Bodh Gaya, monastery groves; Ashoka's edicts record planting banyans and mango groves along roads for shade (3rd century BCE).
- ☐ **GRD-05.6 Mughal gardens**: char bagh, terraces and water; Ram Bagh (Agra), Humayun's Tomb (Delhi), Shalimar and Nishat Bagh (Srinagar), Pinjore (Yadavindra) Gardens (Haryana), the Taj Mahal.
- ☐ **GRD-05.7 Rajput gardens**: palace and lake gardens in Rajasthan, such as Saheliyon-ki-Bari in Udaipur (18th century) and the Kesar Kyari saffron garden in Amber's Maota lake; they blend Mughal water and local courtyards.
- ☐ **GRD-05.8 Kerala homegardens**: multi-storey food forests around houses, with coconut and arecanut at the top, pepper and banana in the middle, and tubers and herbs below; studied by scientists as a model of sustainable agroforestry.
- ☐ **GRD-05.9 Courtyard and tulsi gardens**: the *aangan* with its tulsi platform (*tulsi chaura*), found across Indian homes.
- ☐ **GRD-05.10 Colonial and botanical gardens**: the Indian Botanic Garden at Howrah (1787), Lalbagh in Bengaluru (begun by Hyder Ali and Tipu Sultan, glasshouse 1889), Ooty Botanical Garden (1848); hill-station gardens in Darjeeling and Shimla.
- ☐ **GRD-05.11 Modern Indian gardens**: Amrit Udyan at Rashtrapati Bhavan (renamed from Mughal Gardens in January 2023), Nek Chand's Rock Garden (Chandigarh), Lodhi Garden (Delhi), Brindavan Gardens (Mysuru), the Indira Gandhi Memorial Tulip Garden (Srinagar, 2007).

GRD-06 Garden traditions of the world

Before you start — GRD-06. Each tradition answers the same question differently: should a garden tame nature, copy it or improve on it? Formal gardens use straight lines; landscape gardens hide them.

- ☐ **GRD-06.1 Ancient Egypt and Mesopotamia**: walled gardens with pools; the Hanging Gardens of Babylon, one of the Seven Wonders, have never been found (Stephanie Dalley argues they were at Nineveh, built by Sennacherib).
- ☐ **GRD-06.2 Persian paradise garden**: the walled *bagh* with crossing channels; Pasargadae (6th century BCE) is the oldest known; nine Persian gardens are a UNESCO World Heritage Site (2011).
- ☐ **GRD-06.3 Islamic and Spanish Moorish gardens**: courtyard gardens with water at the centre; the Alhambra and Generalife in Granada.

☐ **GRD-06.4 Chinese scholar gardens**: rocks, water, pavilions and borrowed views designed for poetry and thought; the Classical Gardens of Suzhou (UNESCO).

☐ **GRD-06.5 Japanese gardens**: *karesansui* dry gardens of raked gravel and stone (Ryōan-ji, Kyoto); tea gardens (*roji*) with stepping stones; stroll gardens around ponds (Katsura). Ideas: *wabi-sabi*, borrowed scenery (*shakkei*).

☐ **GRD-06.6 Roman gardens**: peristyle courtyards, topiary and frescoes, preserved at Pompeii.

☐ **GRD-06.7 Medieval monastery gardens**: herb, physic and vegetable beds; the Plan of St Gall (about 820 CE).

☐ **GRD-06.8 Italian Renaissance gardens**: terraces, fountains and statues on hillsides; Villa d'Este at Tivoli.

☐ **GRD-06.9 French formal gardens**: grand geometry and parterres; André Le Nôtre's Versailles (from 1661).

☐ **GRD-06.10 Dutch gardens**: small, neat and full of bulbs; the tulip mania of 1636 to 1637 (whose scale historians now debate); Keukenhof today.

☐ **GRD-06.11 English landscape gardens**: "nature improved": rolling lawns, lakes and clumps of trees by William Kent and Lancelot "Capability" Brown (Stowe, Blenheim).

☐ **GRD-06.12 English cottage gardens**: crowded, informal beds of flowers, herbs and vegetables; refined by Gertrude Jekyll.

☐ **GRD-06.13 African gardens**: homestead gardens, Ethiopian *enset* gardens, Kirstenbosch in Cape Town (1913) for South Africa's unique fynbos flora.

☐ **GRD-06.14 Latin American gardens**: Aztec *chinampas* (floating gardens), Roberto Burle Marx's modernist tropical gardens in Brazil.

☐ **GRD-06.15 Modern and naturalistic gardens**: the New Perennial movement of Piet Oudolf (High Line, New York, 2009); Singapore's Gardens by the Bay (2012) and its "city in nature" plan.

PART 4 — THE PLANT WORLD FOR GARDENERS

About this Part. Enough botany to garden well: how plants are named and grouped, how long they live, what shapes they take, which plants are native to each part of India, and which crops belong to which season.

GRD-07 Names, families, life cycles and plant types

Before you start — GRD-07. Every plant has one scientific (botanical) name in two parts, genus and species, a system made standard by Carl Linnaeus in *Species Plantarum* (1753). Common names change from place to place: "brinjal", "aubergine" and "eggplant" are all *Solanum melongena*. Kew's *Plants of the World Online* is the reference for accepted names.

☐ GRD-07.1 Plant families every gardener meets

☐ **GRD-07.1a Solanaceae (nightshades)**: tomato, potato, brinjal, chilli, capsicum, petunia; share many pests and diseases, so rotate them together.

☐ **GRD-07.1b Fabaceae (legumes)**: peas, beans, lentils, moong, chana, groundnut, also trees like the Indian laburnum (amaltas); fix nitrogen through root nodules.

☐ **GRD-07.1c Brassicaceae (cabbage family)**: cabbage, cauliflower, broccoli, mustard, radish, turnip; cool-season crops in India.

☐ **GRD-07.1d Cucurbitaceae (gourds)**: pumpkin, cucumber, bottle gourd (lauki), bitter gourd (karela), ridge gourd, melons; warm-season climbers.

☐ **GRD-07.1e Poaceae (grasses)**: rice, wheat, maize, millets, sugarcane, bamboo, lawn grasses.

☐ **GRD-07.1f Lamiaceae (mint family)**: mint, tulsi and basil, rosemary, thyme, oregano; square stems and scented leaves.

☐ **GRD-07.1g Apiaceae (carrot family)**: carrot, coriander, cumin, fennel, ajwain; umbrella-shaped flower heads loved by beneficial insects.

☐ **GRD-07.1h Asteraceae (daisy family)**: marigold, sunflower, zinnia, chrysanthemum, lettuce; the largest flowering plant family alongside the orchids.

☐ **GRD-07.1i Amaryllidaceae (onion family, in the broad sense)**: onion, garlic, leek and ornamental bulbs.

☐ **GRD-07.1j Rosaceae (rose family)**: rose, apple, pear, strawberry, peach, almond.

☐ **GRD-07.1k Rutaceae (citrus family)**: lemon, orange, curry leaf, bael.

☐ **GRD-07.1l Zingiberaceae (ginger family)**: ginger, turmeric, cardamom.

☐ GRD-07.2 Life cycles

☐ **GRD-07.2a Annuals**: complete life in one season (marigold, coriander, most vegetables).

☐ **GRD-07.2b Biennials**: grow leaves in year one and flower in year two (carrot, onion and cabbage when grown for seed).

☐ **GRD-07.2c Perennials**: live for many years (curry leaf, roses, fruit trees, asparagus); herbaceous perennials die back and return.

☐ **GRD-07.2d "Grown as annual"**: tropical perennials like chilli and tomato are often grown for one season.

☐ GRD-07.3 Plant types

☐ **GRD-07.3a Trees**: one woody trunk, usually over about 5 m.

☐ **GRD-07.3b Shrubs**: several woody stems, under about 5 m (hibiscus, ixora).

☐ **GRD-07.3c Climbers and creepers**: twining (beans), tendril (gourds), scrambling (bougainvillea), clinging (money plant).

☐ **GRD-07.3d Grasses and bamboos**.

☐ **GRD-07.3e Bulbs, corms, tubers and rhizomes**: onion, gladiolus, potato, ginger; underground food stores.

☐ **GRD-07.3f Ferns, mosses and cycads**: older plant groups that reproduce by spores (ferns, mosses) or seeds in cones (cycads).

☐ **GRD-07.3g Succulents and epiphytes**: water-storers (aloe, cacti) and air-plants that grow on trees (many orchids).

☐ **GRD-07.3h Aquatic plants**: lotus, water lily, water chestnut (singhara).

GRD-08 Native plants of India by region

Before you start — GRD-08. Native plants are adapted to local soil, rain and insects, need less care, and feed local wildlife. The Botanical Survey of India (1890) documents India's flora, which BSI counts at over 21,000 species of flowering plants (check bsi.gov.in for the latest count). Exotic is not bad, but invasive species such as *Lantana camara*, *Prosopis juliflora* (vilayati babool) and water hyacinth damage ecosystems.

☐ **GRD-08.1 Himalaya and the hills**: deodar, rhododendron (buransh), oak (banj), blue pine, Himalayan cherry (padam), primulas; many orchids in the east.

☐ **GRD-08.2 North Indian plains**: neem, peepal, banyan, amaltas, palash (flame of the forest), shisham, arjun, kadamba, harsingar (parijat).

☐ **GRD-08.3 Thar and the dry west**: khejri (Rajasthan's state tree), rohida, ber, kair, phog, aak, desert grasses such as sewan.

☐ **GRD-08.4 Deccan plateau**: tamarind, jamun, karanj (pongamia), sandalwood, mahua, custard apple in the wild.

☐ **GRD-08.5 Western Ghats**: a biodiversity hotspot; many endemic species, wild pepper and cardamom, *Strobilanthes kunthiana* (neelakurinji), which blooms once in about 12 years.

☐ **GRD-08.6 Eastern and north-eastern India**: sal, bamboos, orchids (Arunachal and Sikkim), wild citrus (India's north-east is a centre of citrus diversity), the Assam tea plant.

☐ **GRD-08.7 Coast and islands**: coconut, mangroves (Sundarbans, Bhitarkanika), screwpine (kewda), sea almond (*Terminalia catappa*).

GRD-09 Seasons and crop calendars

Before you start — GRD-09. India has three cropping seasons named after the monsoon cycle, while temperate countries think in spring, summer, autumn and winter. Gardeners in the hills follow a temperate calendar; gardeners in the south can sow most crops nearly year-round. See GRD-25 for a month-by-month calendar.

☐ **GRD-09.1 Kharif (monsoon season)**: sown with the rains, June to July, harvested September to October: rice, maize, millets, moong, groundnut, cotton; garden okra, gourds, beans, brinjal.

☐ **GRD-09.2 Rabi (winter season)**: sown October to November, harvested March to April: wheat, mustard, chana, peas; garden cauliflower, cabbage, carrot, radish, spinach, methi, coriander, winter flowers.

☐ **GRD-09.3 Zaid (short summer season)**: sown February to March with irrigation: watermelon, muskmelon, cucumber, bottle gourd, moong.

☐ **GRD-09.4 Temperate seasons**: spring sowing after the last frost, summer growth, autumn harvest and bulb planting, winter rest; used in the Himalaya and most of Europe and North America.

☐ **GRD-09.5 Day length**: some plants flower by day length (photoperiodism): onion varieties are bred as "short-day" for India's latitudes and "long-day" for northern Europe.

PART 5 — SOIL, WATER, LIGHT AND CLIMATE

About this Part. The four things every plant needs from its place. Get these right and most problems never appear.

GRD-10 Soil

Before you start — GRD-10. ICAR classifies India's soils into eight major groups (alluvial, black, red and yellow, laterite, arid, saline, peaty and forest soils); see the Geography index (GEO) of the Curriculum Index for the soil map. The Soil Health Card scheme (2015) gives farmers free soil tests and advice. **See also:** GRD-00.03.

- ☐ **GRD-10.1 Texture**: sand (drains fast, holds little), silt, clay (holds water and nutrients, drains slowly), loam (a balance; the gardener's ideal). Test: wet a pinch and rub; gritty is sandy, sticky is clay.
- ☐ **GRD-10.2 Structure**: how particles clump into crumbs; organic matter and roots build it, compaction and heavy tilling break it.
- ☐ **GRD-10.3 pH**: acidity scale from 0 to 14; most vegetables like 6.0 to 7.5; blueberries, tea and azaleas like acid soil; much of Rajasthan's soil is alkaline.
- ☐ **GRD-10.4 Nutrients**: the big three NPK (nitrogen for leaves, phosphorus for roots and flowers, potassium for overall strength), secondary (calcium, magnesium, sulphur) and micronutrients (iron, zinc, boron, manganese, copper, molybdenum).
- ☐ **GRD-10.5 Indian soil groups for gardeners**: alluvial (Indo-Gangetic plains, fertile), black or regur (Deccan, holds water, cracks when dry; good for cotton), red and laterite (south and east, often acidic and low in nutrients; add compost), arid desert soils (sandy, low organic matter; mulch heavily).
- ☐ **GRD-10.6 Soil testing**: home pH strips, state soil-testing labs, Soil Health Card portal (soilhealth.dac.gov.in).

GRD-11 Compost, mulch and fertilisers

Before you start — GRD-11. Compost is the gardener's gold: decomposed organic matter that feeds soil life, improves structure and holds water. A balanced heap mixes "greens" (kitchen scraps, fresh leaves: nitrogen) and "browns" (dry leaves, paper, straw: carbon).

- ☐ **GRD-11.1 Hot composting**: a large, turned heap that heats to about 55 to 65 °C, killing many weed seeds; ready in weeks to a few months. The Indore method was developed by Albert Howard and Yeshwant Wad at Indore in the 1920s and 1930s, and Howard's *An Agricultural Testament* (1940) helped start the organic movement.
- ☐ **GRD-11.2 Cold composting**: a slow pile left alone for six months to a year.
- ☐ **GRD-11.3 Vermicompost**: compost made by earthworms (usually red wigglers, *Eisenia fetida*); fine, rich castings.
- ☐ **GRD-11.4 Bokashi**: fermenting food waste with inoculated bran in a sealed bucket, then burying it; developed in Japan.
- ☐ **GRD-11.5 Leaf mould**: decomposed dry leaves; an excellent soil conditioner.
- ☐ **GRD-11.6 Home and community composting in India**: khamba (terracotta stacks), aerobic bins, Bengaluru's and Pune's resident-welfare composting.
- ☐ **GRD-11.7 Mulch**: a layer of straw, dry leaves, wood chips or cardboard on the soil; keeps moisture in, keeps weeds down and cools roots in Indian summers.
- ☐ **GRD-11.8 Organic inputs**: cow-dung manure, neem cake, bone meal, wood ash (potassium), *jeevamrut*, *panchagavya* and *kunapajala* (traditional ferments; evidence is still limited), seaweed extract.
- ☐ **GRD-11.9 Mineral fertilisers**: urea, DAP and NPK mixes; fast but easy to overuse; never put fertiliser on dry soil or touching roots.
- ☐ **GRD-11.10 Green manure and cover crops**: sunn hemp, dhaincha, cowpea grown and turned into the soil.

GRD-12 Water, light and climate

Before you start — GRD-12. Most plant deaths at home come from too much or too little water, not from pests. Water

deeply and less often, so roots grow down.

☐ GRD-12.1 Water

☐ **GRD-12.1a Watering by hand**: early morning is best; water the soil, not the leaves.

☐ **GRD-12.1b Drip irrigation**: water dripped at the root zone; saves 30 to 60 per cent of water compared with flood irrigation, by many Indian studies; Israel's Netafim (1965) made it commercial. India's PM Krishi Sinchayee Yojana promotes "more crop per drop".

☐ **GRD-12.1c Wicking and self-watering pots**: a wick or reservoir under the soil.

☐ **GRD-12.1d Clay-pot irrigation**: buried unglazed pots (*olla* or *matka* irrigation) that seep water slowly; an old technique of many dry lands.

☐ **GRD-12.1e Rainwater harvesting**: barrels, recharge pits and the traditional tankas and johads of Rajasthan.

☐ **GRD-12.1f Signs of trouble**: wilting with wet soil and yellow leaves usually means overwatering and root rot; crisp, dry leaf edges mean underwatering or heat.

☐ GRD-12.2 Light

☐ **GRD-12.2a Full sun**: 6 hours or more of direct sun (most vegetables and flowers).

☐ **GRD-12.2b Partial sun or shade**: 3 to 6 hours (leafy greens, many herbs).

☐ **GRD-12.2c Full shade**: under 3 hours (ferns, many indoor plants).

☐ **GRD-12.2d Indoor light**: "bright indirect" means near a window but out of harsh direct rays; LED grow lights help.

☐ GRD-12.3 Climate

☐ **GRD-12.3a USDA hardiness zones**: a map of the coldest winter temperature in 13 zones; used worldwide in seed catalogues but built for cold climates.

☐ **GRD-12.3b RHS hardiness ratings**: H1a to H7, for the UK.

☐ **GRD-12.3c India's agro-climatic zones**: the Planning Commission defined 15 agro-climatic regions (1989), and ICAR's National Agricultural Research Project uses 127 agro-climatic zones; your state agricultural university's crop calendar is the best guide.

☐ **GRD-12.3d Microclimates**: a warm south-facing wall, a windy terrace or a shady courtyard can differ from the city's weather.

☐ **GRD-12.3e Climate change and gardening**: earlier springs, heatwaves and erratic rain; responses include mulch, shade nets, heat-tolerant varieties and rainwater storage.

PART 6 — TECHNIQUES

About this Part. The gardener's skills, from the first seed to the saved seed. Each technique has a one-line "how" and a link to a fuller website page.

GRD-13 Sowing and propagation

Before you start — GRD-13. Plants make new plants in two ways: sexually, by seed (children differ from parents), and vegetatively, from a piece of the parent (clones that keep the variety true). Mangoes, roses and most fruit trees are propagated vegetatively for this reason.

- ☐ **GRD-13.1 Direct sowing**: seeds sown where they will grow (radish, carrot, coriander, beans, gourds); the rule of thumb is to sow at a depth about two to three times the seed's width.
- ☐ **GRD-13.2 Nursery and seed trays**: sow small seeds in trays or a nursery bed (tomato, chilli, brinjal, cabbage, flowers), then transplant.
- ☐ **GRD-13.3 Transplanting and hardening off**: move seedlings at four to six true leaves, in the evening, with a ball of soil; give them a few days of gradual sun first.
- ☐ **GRD-13.4 Seed treatment**: soaking hard seeds, scarification (nicking the coat), stratification (cold treatment for temperate seeds), and *beejamrut* or fungicide coatings.
- ☐ **GRD-13.5 Cuttings**: stem cuttings (rose, hibiscus, money plant, bougainvillea), leaf cuttings (succulents, snake plant), root cuttings; rooting hormone helps woody cuttings.
- ☐ **GRD-13.6 Layering**: rooting a branch while still on the parent; simple layering (jasmine), air layering or *gootee* (litchi, pomegranate, guava).
- ☐ **GRD-13.7 Grafting**: joining a shoot (scion) to a rooted plant (rootstock); mango's veneer and soft-wood grafting, apple's whip-and-tongue, cleft grafting.
- ☐ **GRD-13.8 Budding**: grafting a single bud; T-budding in roses and citrus, patch budding in ber.
- ☐ **GRD-13.9 Division**: splitting clumps of perennials, grasses and bulbs (lemongrass, aloe, spider plant).
- ☐ **GRD-13.10 Runners, suckers and offsets**: strawberry runners, banana suckers, pineapple crowns.
- ☐ **GRD-13.11 Tissue culture**: growing plants from tiny pieces in sterile jars; India produces tissue-culture bananas and orchids at scale; certified under the government's National Certification System for Tissue Culture Raised Plants.

GRD-14 Care, training and planning

Before you start — GRD-14. Good gardens are planned before they are planted: what grows next to what, where it grew last year, and what you will eat when.

- ☐ **GRD-14.1 Pruning**: removing dead, crossing or crowded branches; heading back (bushier) versus thinning (airier); prune most trees in their dormant or post-harvest season.
- ☐ **GRD-14.2 Pinching and deadheading**: pinching tips for bushy basil and marigold; removing spent flowers for more blooms.
- ☐ **GRD-14.3 Training, staking and trellising**: stakes for tomatoes, *pandal* (bamboo trellis) for gourds, espalier for fruit trees on walls.
- ☐ **GRD-14.4 Thinning**: removing extra seedlings so the rest have room (carrots, radish, beet).
- ☐ **GRD-14.5 Weeding**: by hand, hoe or mulch; some "weeds" are edible (bathua, chaulai) or medicinal.
- ☐ **GRD-14.6 Companion planting**: plants that help each other; marigold with tomato (traditional, with some evidence against root-knot nematodes), the "three sisters" (maize, beans, squash), basil near tomato. Many companion claims are folklore; test them.
- ☐ **GRD-14.7 Intercropping and mixed cropping**: two or more crops in one plot; India's traditional *baranaja* ("twelve grains") of Uttarakhand.

☐ **GRD-14.8 Crop rotation**: change the family grown in a bed each season (legumes, then leafy brassicas, then fruiting nightshades, then roots) to break pest cycles and balance nutrients.

☐ **GRD-14.9 Succession sowing**: sow small batches every two to three weeks (coriander, radish) for a steady harvest.

☐ **GRD-14.10 Harvesting**: pick greens young, okra and beans every two days, tomatoes at the colour break; harvest in the cool of the morning.

☐ **GRD-14.11 Seed saving**: save seed from open-pollinated, healthy plants; hybrids (F1) do not breed true. Community seed banks such as those of the Deccan Development Society and Navdanya keep heirloom varieties alive. India's Protection of Plant Varieties and Farmers' Rights Act (2001) protects farmers' right to save and share seed.

PART 7 — PLANT HEALTH

About this Part. What goes wrong and how to respond, using the ladder of integrated pest management (IPM): prevent first, watch closely, act gently, and use stronger measures only when needed.

GRD-15 Pests, diseases and integrated pest management

Before you start — GRD-15. IPM is the approach recommended by the FAO and by India's Directorate of Plant Protection. Most pests have natural enemies: ladybirds eat aphids, spiders and birds eat caterpillars. Sprays that kill everything also kill the helpers. **Caution:** even organic sprays should be used carefully, away from bees in flower and out of children's and pets' reach.

☐ GRD-15.1 Common pests

☐ **GRD-15.1a Aphids**: tiny soft insects on new shoots; spray water, neem oil soap, encourage ladybirds.

☐ **GRD-15.1b Whiteflies and mealybugs**: sap-suckers that leave sticky honeydew and sooty mould; yellow sticky traps, neem oil.

☐ **GRD-15.1c Mites**: tiny spiders causing speckled, dusty leaves in dry heat; mist and wash leaves.

☐ **GRD-15.1d Caterpillars and borers**: fruit and shoot borer in brinjal, fruit borer in tomato; hand-pick, pheromone traps, *Bacillus thuringiensis* (Bt) sprays.

☐ **GRD-15.1e Fruit flies**: maggots in gourds and mangoes; bag fruits, methyl eugenol or cue-lure traps.

☐ **GRD-15.1f Snails and slugs**: monsoon pests; hand-pick at night, barriers.

☐ **GRD-15.1g Root-knot nematodes**: microscopic worms making galls on roots; rotation, marigold, neem cake.

☐ **GRD-15.1h Birds, squirrels, monkeys and rats**: nets and cages; share a little.

☐ GRD-15.2 Common diseases

☐ **GRD-15.2a Damping-off**: seedlings collapse at the soil line; sterile mix, less water, good air.

☐ **GRD-15.2b Powdery mildew**: white powder on leaves of gourds and roses; airflow, diluted milk or baking-soda sprays (evidence mixed), sulphur.

☐ **GRD-15.2c Leaf spots and blights**: early and late blight of tomato and potato; late blight caused the Irish famine of 1845 to 1849.

☐ **GRD-15.2d Wilts and root rots**: *Fusarium*, bacterial wilt; resistant varieties, *Trichoderma* biocontrol, good drainage.

☐ **GRD-15.2e Viruses**: leaf curl in chilli and tomato, yellow vein mosaic in okra, spread by whiteflies; pull out infected plants, control the insect.

☐ GRD-15.3 The IPM ladder

☐ **GRD-15.3a Prevent**: healthy soil, resistant varieties, rotation, spacing, clean tools.

☐ **GRD-15.3b Monitor**: check leaves weekly, especially undersides; use sticky and pheromone traps.

☐ **GRD-15.3c Physical and cultural control**: hand-picking, pruning, nets, water jets.

☐ **GRD-15.3d Biological control**: ladybirds, lacewings, *Trichogramma* wasps, Bt, *Trichoderma*.

☐ **GRD-15.3e Botanical sprays**: neem (azadirachtin), garlic-chilli sprays; still use with care.

☐ **GRD-15.3f Chemical control as a last resort**: only registered products from the Central Insecticides Board list, at the label dose.

GRD-16 Nutrient deficiencies (symptoms)

Before you start — GRD-16. Where the yellowing appears tells you a lot. Nutrients that move within the plant (N, P, K, Mg) show first on **old, lower leaves**; nutrients that do not move (Fe, Ca, B, S) show first on **young, upper leaves**. Wrong pH can lock nutrients away even when they are present.

☐ **GRD-16.1 Nitrogen (N)**: old leaves turn evenly pale yellow; slow growth. Fix: compost, manure, legumes.

☐ **GRD-16.2 Phosphorus (P)**: dull, dark or purplish leaves, poor roots and flowering. Fix: bone meal, rock phosphate, compost.

☐ **GRD-16.3 Potassium (K)**: scorched, brown leaf edges on old leaves; weak stems. Fix: wood ash, banana-peel compost, potash.

☐ **GRD-16.4 Magnesium (Mg)**: yellow between veins on old leaves, veins stay green. Fix: Epsom salt (magnesium sulphate) in small doses.

☐ **GRD-16.5 Iron (Fe)**: yellow between veins on young leaves; common in alkaline soil. Fix: lower pH, chelated iron.

☐ **GRD-16.6 Calcium (Ca)**: blossom-end rot in tomatoes (black base of fruit), tip burn; usually caused by uneven watering. Fix: steady watering, lime or gypsum if soil is low.

☐ **GRD-16.7 Boron (B)**: hollow stems in cauliflower, cracked fruit, deformed tips. Fix: a pinch of borax, carefully; too much is toxic.

☐ **GRD-16.8 Zinc (Zn)**: small, narrow leaves (little leaf); India's most widespread micronutrient deficiency in soils. Fix: zinc sulphate as advised by a soil test.

PART 8 — REGISTERS

About this Part. Three registers: 30 plant cards for Indian and international gardeners, a year-wise timeline of gardening history, and the great shows and awards with their latest results. Facts are a sourced selection; the full plant database grows on the website.

GRD-17 Plant cards (register of 30 garden plants)

Before you start — GRD-17. The brief's full card has thirteen fields. To fit the page, each plant appears twice: Card A (identity) and Card B (care). Days are typical from sowing or planting in Indian conditions; local varieties differ. Sun: F full, P part, S shade. Water: L low, M medium, H high. **Cautions** flag plants toxic to pets or people.

☐ GRD-17.1 Card A: identity

Plant (English)	Hindi	Botanical name	Family	Type	Season (India)
Tomato	Tamatar	*Solanum lycopersicum*	Solanaceae	Annual (perennial in tropics)	Rabi; also kharif in south
Brinjal	Baingan	*Solanum melongena*	Solanaceae	Short perennial grown as annual	Kharif and rabi
Chilli	Mirch	*Capsicum annuum*	Solanaceae	Short perennial	Kharif and rabi
Okra	Bhindi	*Abelmoschus esculentus*	Malvaceae	Annual	Zaid and kharif
Spinach	Palak	*Spinacia oleracea*	Amaranthaceae	Annual	Rabi
Fenugreek	Methi	*Trigonella foenum-graecum*	Fabaceae	Annual	Rabi
Coriander	Dhaniya	*Coriandrum sativum*	Apiaceae	Annual	Rabi
Radish	Mooli	*Raphanus sativus*	Brassicaceae	Annual	Rabi
Carrot	Gajar	*Daucus carota* subsp. *sativus*	Apiaceae	Biennial grown as annual	Rabi
Cauliflower	Phool gobhi	*Brassica oleracea* (Botrytis group)	Brassicaceae	Annual	Rabi
Peas	Matar	*Pisum sativum*	Fabaceae	Annual	Rabi
Bottle gourd	Lauki	*Lagenaria siceraria*	Cucurbitaceae	Annual climber	Zaid and kharif
Bitter gourd	Karela	*Momordica charantia*	Cucurbitaceae	Annual climber	Zaid and kharif
Cucumber	Kheera	*Cucumis sativus*	Cucurbitaceae	Annual climber	Zaid and kharif
Onion	Pyaz	*Allium cepa*	Amaryllidaceae	Biennial grown as annual	Rabi (also kharif)
Garlic	Lahsun	*Allium sativum*	Amaryllidaceae	Bulb	Rabi
Mint	Pudina	*Mentha* spp.	Lamiaceae	Perennial	Year-round
Holy basil	Tulsi	*Ocimum tenuiflorum*	Lamiaceae	Short perennial	Year-round (dislikes frost)
Curry leaf	Kadi patta	*Murraya koenigii*	Rutaceae	Shrub or small tree	Year-round
Lemongrass	Nimbu ghaas, hari chai	*Cymbopogon citratus*	Poaceae	Perennial grass	Year-round
Aloe vera	Ghritkumari	*Aloe vera*	Asphodelaceae	Succulent perennial	Year-round
Moringa	Sahjan, saijan	*Moringa oleifera*	Moringaceae	Tree	Year-round
Papaya	Papita	*Carica papaya*	Caricaceae	Fast tree-like herb	Year-round (frost-tender)
Lemon	Nimbu	*Citrus limon*	Rutaceae	Shrub or tree	Year-round
Marigold	Genda	*Tagetes erecta*	Asteraceae	Annual	All three seasons
Rose	Gulab	*Rosa* spp.	Rosaceae	Shrub	Plant in winter (plains)
Hibiscus	Gudhal	*Hibiscus rosa-sinensis*	Malvaceae	Shrub	Year-round
Jasmine (Arabian)	Mogra	*Jasminum sambac*	Oleaceae	Shrub	Flowers in summer

Money plant	Money plant	*Epipremnum aureum* (pothos)	Araceae	Climber	Year-round, indoors
Snake plant	Sansevieria	*Dracaena trifasciata*	Asparagaceae	Succulent perennial	Year-round, indoors

☐ GRD-17.2 Card B: care

Plant	Sun · Water · Soil	Propagation	Days to harvest or flower	Uses	Cautions
Tomato	F · M · loam, pH 6–7	Seed, transplant	60–90 after transplant	Fruit, sauces	Leaves toxic to pets
Brinjal	F · M · loam	Seed, transplant	60–80 after transplant	Bharta, curries	Leaves toxic if eaten
Chilli	F · M · well-drained	Seed, transplant	70–90 after transplant	Spice, pickles	Wash hands; keep from eyes
Okra	F · M · loam	Direct seed (soak overnight)	45–60	Vegetable	Hairy pods can irritate skin
Spinach	F/P · M · rich loam	Direct seed	30–45 (cut again)	Greens	High oxalate
Fenugreek	F/P · M · any	Direct seed	20–30 for leaves	Greens, seeds as spice	None notable
Coriander	F/P · M · well-drained	Direct seed (split seeds)	30–40 for leaves	Herb, seed spice	Bolts in heat
Radish	F · M · loose	Direct seed	25–45	Root, leaves, pods	None notable
Carrot	F · M · deep sandy loam	Direct seed	75–90	Root	None notable
Cauliflower	F · M · rich, boron needed	Seed, transplant	60–100 after transplant	Vegetable	None notable
Peas	F · M · any	Direct seed	60–75	Pods, shoots	None notable
Bottle gourd	F · M · rich	Direct seed	55–70	Vegetable, juice	Bitter fruit can be toxic: discard
Bitter gourd	F · M · rich	Direct seed	55–65	Vegetable	Avoid in pregnancy in large amounts
Cucumber	F · M/H · rich	Direct seed	45–60	Salad	Bitter fruit: discard
Onion	F · M · loose	Seed nursery or sets	120–150 (bulb)	Bulb, greens	Toxic to cats and dogs
Garlic	F · L/M · loose	Cloves	130–150	Bulb, spice	Toxic to cats and dogs
Mint	P · H · moist	Cuttings, runners	30 from cuttings	Herb, chutney	Spreads aggressively: use pots
Holy basil	F · M · well-drained	Seed, cuttings	40–60 for leaves	Worship, tea	Medicinal claims need care
Curry leaf	F · M · well-drained	Seed (fresh), suckers	1–2 years to bushy plant	Tadka, chutneys	Slow when young
Lemongrass	F · M · any	Division	90–120	Tea, Thai curry	Mild irritant to pets
Aloe vera	F/P · L · sandy	Offsets	2–3 years to mature	Gel for minor burns	Latex is a laxative; toxic to pets
Moringa	F · L · well-drained	Seed, big cuttings	Pods in 6–8 months	Leaves, pods (drumsticks)	None notable in food amounts
Papaya	F · M · well-drained	Seed	9–11 months to fruit	Fruit, raw papaya curry	Latex irritant; root rot if waterlogged
Lemon	F · M · well-drained	Air layer, budding, seed	2–3 years from layer	Juice, pickle	Thorns
Marigold	F · L/M · any	Seed, cuttings	45–60 to flower	Garlands, pest trap crop	Can irritate skin
Rose	F · M · rich loam	Cuttings, budding	45–60 per flush	Flowers, gulkand	Thorns
Hibiscus	F · M · rich	Cuttings, air layer	Flowers most of year	Flowers, hair oil	Mealybugs love it
Jasmine (Arabian)	F · M · loam	Cuttings, layering	Flowers in first summer	Garlands, perfume	Non-toxic
Money plant	P/S · M · well-drained	Cuttings in water	Fast growth	Indoor decor	Toxic to pets and children if chewed
Snake plant	P/S · L · sandy	Division, leaf cuttings	Slow	Indoor decor	Toxic to pets if chewed

GRD-18 The Gardening Timeline (year-wise register)

Year	Event	Where	Why it matters
c. 9500 BCE	First domesticated wheat, barley and lentils	Fertile Crescent	Start of farming
c. 7000 BCE	Barley, wheat and cattle at Mehrgarh	Balochistan, South Asia	South Asia's earliest farming village
c. 1400 BCE	Tomb paintings of walled gardens with pools	Egypt	Earliest pictures of pleasure gardens
6th century BCE	Garden of Cyrus the Great at Pasargadae	Persia	Oldest known char bagh plan
3rd century BCE	Ashoka's edicts record roadside trees and wells	India	Public planting as state duty
c. 300 BCE	Theophrastus, *Enquiry into Plants*	Greece	"Father of botany"
6th century CE	Varahamihira's *Brihat Samhita*, chapter on trees	India	Early Indian plant-care text
c. 820 CE	Plan of St Gall shows monastery herb gardens	Switzerland	Model of medieval gardens
c. 10th century	Surapala's *Vrikshayurveda*	India	Sanskrit science of plant life
1339	Saihō-ji garden remodelled by Musō Soseki	Kyoto, Japan	Root of Zen garden design
1528	Babur lays out the Aram Bagh	Agra, India	First Mughal garden
1545	Orto Botanico di Padova	Italy	Oldest university botanical garden still on its site
1620s	Shalimar Bagh	Srinagar	Mughal terraced water garden
1661	Le Nôtre begins Versailles gardens	France	Peak of formal style
1753	Linnaeus, *Species Plantarum*	Sweden	Two-part plant names
1759	Royal Botanic Gardens, Kew, founded	England	World centre of plant science
1787	Indian Botanic Garden founded by Robert Kyd	Howrah, Kolkata	Asia's great colonial garden; the Great Banyan
1804	Royal Horticultural Society founded	London	World's leading gardening charity
1829	Nathaniel Ward's sealed glass case	London	Plants survive long sea voyages
1848	Botanical Garden, Ooty	Tamil Nadu	Hill-station garden
1866	Mendel's pea experiments published	Brno	Basis of genetics and plant breeding
1890	Botanical Survey of India	India	National plant survey
1913	First Chelsea Flower Show	London	The world's most famous garden show
1924 to 1931	Indore method of composting	Indore	Seed of the organic movement
1940	Albert Howard, *An Agricultural Testament*	UK	Organic farming classic
1960s	Green Revolution: high-yield wheat and rice	India, Mexico, Philippines	Food security; later soil and water costs
1962	Rachel Carson, *Silent Spring*	USA	Pesticide warning; birth of environmentalism
1975	Fukuoka, *The One-Straw Revolution*	Japan	Natural farming
1976	Nek Chand's Rock Garden opens	Chandigarh	Garden of recycled art
1978	*Permaculture One* (Mollison and Holmgren)	Australia	Permaculture named
2001	PPV&FR Act	India	Farmers' seed rights
2009	High Line park opens	New York	Naturalistic urban planting
2016	Sikkim declared fully organic	India	First organic state

2023	Mughal Gardens renamed Amrit Udyan	New Delhi	Rashtrapati Bhavan gardens
2024	National Mission on Natural Farming approved	India	Government push for natural farming
2026	Chelsea's top prize: Sarah Eberle's CPRE garden "On the Edge"	London	Latest Chelsea Garden of the Year

PART 9 — HISTORY

About this Part. The story of gardening in seven chapters, each a website page. The timeline (GRD-18) gives the dates.

GRD-19 From the first farmers to city rooftops

Before you start — GRD-19. Gardening's history is also a history of plants moving: chilli, potato, tomato and maize reached India only after 1498 with Portuguese and other traders, and India's mango, rice, sugarcane, cotton and black pepper travelled the other way. **See also:** COK (the history of food).

- ☐ **GRD-19.1 The first farmers**: the Neolithic revolution in many cradles (GRD-00.01); Mehrgarh and the Indus cities' crops.
- ☐ **GRD-19.2 Ancient gardens**: Egypt, Mesopotamia and the Hanging Gardens (location unknown), Persia's paradise gardens, the parks of Ashoka, Greek and Roman gardens, China's imperial parks.
- ☐ **GRD-19.3 Medieval gardens**: monastery herb and physic gardens; Islamic Spain; Chola and Vijayanagara temple gardens; Japanese Zen gardens.
- ☐ **GRD-19.4 Botanical gardens**: Pisa (1544) and Padua (1545) as the first university gardens; Kew (1759); Kolkata (1787); Peradeniya (Sri Lanka, 1821); Singapore Botanic Gardens (1859, UNESCO 2015). Today Botanic Gardens Conservation International (BGCI) counts more than 3,500 botanic gardens and arboreta worldwide (see bgci.org).
- ☐ **GRD-19.5 The plant hunters and empire**: collectors such as Joseph Hooker in Sikkim (1848 to 1850), Robert Fortune, who smuggled tea plants out of China for the East India Company (1848 to 1851), and Frank Kingdon-Ward in the eastern Himalaya; rubber seeds taken from Brazil to Kew (1876) and on to Asia. Gardens were also tools of empire, and the knowledge of local people was rarely credited.
- ☐ **GRD-19.6 The Green Revolution and after**: high-yield wheat and rice, fertiliser and irrigation from the 1960s (Norman Borlaug, M. S. Swaminathan) ended famine fears in India; later came soil fatigue, groundwater loss and the organic and natural-farming movements.
- ☐ **GRD-19.7 Gardening today**: urban and terrace gardening, community gardens, social-media gardeners, seed libraries, climate-wise gardening and the "grow your own" boom of the 2020 pandemic lockdowns.

PART 10 — PEOPLE, SHOWS AND AWARDS

About this Part. The people who shaped gardening, and the shows and honours that celebrate it. Latest results were checked on 3 November 2026.

GRD-20 Gardeners, botanists and designers

Before you start — GRD-20. Gardening's heroes include scientists, designers, writers and farmers who never had a title. This list balances India and the world.

☐ GRD-20.1 India

- ☐ **GRD-20.1a M. S. Swaminathan (1925 to 2023)**: geneticist who led the Green Revolution in India; World Food Prize 1987; Bharat Ratna 2024 (posthumous).
- ☐ **GRD-20.1b Janaki Ammal (1897 to 1984)**: cytogeneticist who bred sweeter sugarcane at Coimbatore and studied Indian plants; reorganised the Botanical Survey of India in the 1950s at Nehru's invitation; Padma Shri 1977; a magnolia is named after her.
- ☐ **GRD-20.1c Saalumarada Thimmakka (died November 2025, aged about 114)**: planted and cared for some 385 banyan trees along a Karnataka highway; Padma Shri 2019.
- ☐ **GRD-20.1d Tulasi Gowda (died December 2024)**: Halakki tribal elder of Karnataka, known as the "encyclopedia of the forest" for her nursery knowledge; Padma Shri 2020 (presented 2021).
- ☐ **GRD-20.1e Rahibai Soma Popere**: the "Seed Mother" of Maharashtra, who saves indigenous seed varieties; Padma Shri 2020.
- ☐ **GRD-20.1f Kamala Pujari (died 2024)**: Odisha farmer who conserved hundreds of paddy varieties; Padma Shri 2019.
- ☐ **GRD-20.1g Nek Chand (1924 to 2015)**: road inspector who secretly built the Rock Garden of Chandigarh from waste.
- ☐ **GRD-20.1h Subhash Palekar**: promoter of zero-budget natural farming; Padma Shri 2016.
- ☐ **GRD-20.1i Vandana Shiva**: founder of Navdanya seed banks (1987) and critic of industrial agriculture; her views are influential and also debated.
- ☐ **GRD-20.1j Hyder Ali and Tipu Sultan**: began Lalbagh in Bengaluru in the 18th century, bringing plants from many countries.

☐ GRD-20.2 The world

- ☐ **GRD-20.2a Theophrastus (c. 371 to 287 BCE)**: Aristotle's student, the "father of botany".
- ☐ **GRD-20.2b Carl Linnaeus (1707 to 1778)**: gave plants their two-part names.
- ☐ **GRD-20.2c Gregor Mendel (1822 to 1884)**: discovered the laws of heredity with garden peas.
- ☐ **GRD-20.2d André Le Nôtre (1613 to 1700)**: designer of Versailles.
- ☐ **GRD-20.2e Lancelot "Capability" Brown (1716 to 1783)**: shaped some 170 English landscape parks.
- ☐ **GRD-20.2f Gertrude Jekyll (1843 to 1932)**: painter-gardener of colour-planned borders, often with the architect Edwin Lutyens (who also designed New Delhi and the Mughal-style garden at Rashtrapati Bhavan).
- ☐ **GRD-20.2g Joseph Hooker (1817 to 1911)**: explored Sikkim's rhododendrons and directed Kew.
- ☐ **GRD-20.2h Albert and Gabrielle Howard**: developed the Indore method of composting in India with Indian colleagues.
- ☐ **GRD-20.2i Masanobu Fukuoka (1913 to 2008)**: Japanese farmer-philosopher of natural farming.
- ☐ **GRD-20.2j Roberto Burle Marx (1909 to 1994)**: Brazilian modernist landscape designer.
- ☐ **GRD-20.2k Piet Oudolf (born 1944)**: Dutch leader of naturalistic perennial planting.
- ☐ **GRD-20.2l Wangari Maathai (1940 to 2011)**: founded Kenya's Green Belt Movement (1977), which planted tens of millions of trees; Nobel Peace Prize 2004.
- ☐ **GRD-20.2m Gardening voices**: David Attenborough's *The Private Life of Plants* (1995); Monty Don (*Gardeners' World*); Charles Dowding (no-dig); Alan Titchmarsh.

GRD-21 Shows and awards (register)

Before you start — GRD-21. Gardening's biggest honours come from shows and horticultural societies. In India, the great flower shows are public festivals.

Show or award	Body	Since	What it honours	Latest (checked 3 Nov 2026)	Note
RHS Chelsea Flower Show	Royal Horticultural Society	1913	Show gardens and plant exhibits; Gold, Silver-Gilt, Silver and Bronze medals	2026 Garden of the Year (the renamed Best in Show): Sarah Eberle, CPRE "On the Edge"	Held each May in London
RHS Chelsea Plant of the Year	RHS	2010	The best new plant launched at Chelsea	2026: *Hosta* 'Red Ninja', bred by Ninja Kramer	Public and expert judging
RHS Award of Garden Merit (AGM)	RHS	1922	Plants proven reliable in UK gardens after trials	Updated list on rhs.org.uk	A buyer's guide
RHS Victoria Medal of Honour	RHS	1897	Lifetime service to horticulture	Announced yearly	Only 63 holders at a time, for the years of Queen Victoria's reign
Lalbagh Flower Show	Mysore Horticultural Society and Karnataka Horticulture Department	1912 (society)	A themed flower display in the Glass House	Held every January and August	Around Republic and Independence Days
Amrit Udyan, Udyan Utsav	Rashtrapati Bhavan	Opened yearly for decades	The President's gardens opened to all	Opens each spring (dates on rashtrapatibhavan.gov.in)	Renamed from Mughal Gardens in 2023

PART 11 — IDEAS: PHILOSOPHY, PSYCHOLOGY AND SOCIETY

About this Part. The philosophy and psychology angles of the brief. These are ideas to think with, presented fairly, with the evidence for each.

GRD-22 Why gardening heals

Before you start — GRD-22. The evidence is promising but still growing (GRD-00.06). Good studies compare gardeners with similar non-gardeners and measure outcomes over time.

- ☐ **GRD-22.1 The evidence**: meta-analysis of 22 studies (Soga and colleagues, 2017); allotment gardeners' wellbeing (Van den Berg and colleagues, 2010); community-garden trials in the USA (Litt and colleagues, *Lancet Planetary Health*, 2023: more fibre, more activity, less stress).
- ☐ **GRD-22.2 Theories**: biophilia (E. O. Wilson, 1984: an inborn love of living things), attention restoration (Kaplan and Kaplan), stress reduction (Ulrich).
- ☐ **GRD-22.3 Horticultural therapy and social prescribing**: the NHS in England prescribes gardening and nature activities; India's horticultural therapy is growing in hospitals and special schools.
- ☐ **GRD-22.4 Soil microbes and mood**: early research on the soil bacterium *Mycobacterium vaccae* in mice (Lowry and colleagues, 2007); a fascinating lead, not proof for humans.

GRD-23 Gardens in thought, faith and society

- ☐ **GRD-23.1 Gardens in religion**: Eden and the Qur'anic *jannah* (garden); the Bodhi tree; Krishna's Vrindavan groves; tulsi worship; Sikh *gurdwara* gardens; Japanese Zen gardens as meditation.
- ☐ **GRD-23.2 Gardens in poetry and art**: Kalidasa's gardens in *Abhijnanashakuntalam*; Persian poets Hafez and Saadi (*Gulistan*, "The Rose Garden", 1258); Tagore's *The Gardener* (1913); Andrew Marvell's "The Garden"; Monet's Giverny; Frances Hodgson Burnett's *The Secret Garden* (1911).
- ☐ **GRD-23.3 The philosophy of patience and seasons**: Voltaire's "we must cultivate our garden" (*Candide*, 1759); the *Bhagavad Gita*'s work without attachment to fruit; gardening as a lesson that growth takes time.
- ☐ **GRD-23.4 Food security**: FAO's work on home gardens; India's nutrition gardens; Cuba's urban farms after 1990.
- ☐ **GRD-23.5 Climate and biodiversity**: gardens as habitat corridors, carbon in soil, the pollinator crisis, peat-free composts, native planting and the fight against invasives.
- ☐ **GRD-23.6 Justice and access**: who gets land, water and green space; the role of women farmers; credit for indigenous knowledge (Nagoya Protocol, 2010; India's Biological Diversity Act, 2002).

PART 12 — THE PRACTICAL GUIDE

About this Part. Hands-on pages for a first garden. Always start small.

GRD-24 A first garden in 30 days

☐ **GRD-24.1 Days 1 to 3, look**: watch where the sun falls and for how many hours; choose 3 to 5 crops you love to eat.

☐ **GRD-24.2 Days 4 to 7, prepare**: get containers with drainage holes or dig a small bed; make potting mix (soil, compost, coco peat).

☐ **GRD-24.3 Days 8 to 10, sow**: methi, coriander, spinach and radish directly; buy a few healthy seedlings (chilli, tomato) from a nursery.

☐ **GRD-24.4 Days 11 to 20, care**: water when the top 2 cm is dry; mulch; check leaves every day; start a compost bucket.

☐ **GRD-24.5 Days 21 to 30, harvest and note**: pick microgreens and methi; write what worked in a garden diary; plan the next sowing.

GRD-25 A month-by-month calendar for India

Before you start — GRD-25. A simplified guide; follow your state agricultural university's calendar for exact varieties and dates.

Month	North Indian plains	Hills (temperate)	South India and coast	Flowers	Tasks
January	Harvest winter crops; sow tomato and brinjal nursery under cover	Rest; prune fruit trees	Sow most vegetables	Winter annuals bloom	Protect from frost
February	Sow summer gourds, okra, cucumber	Prepare beds	Sow gourds, okra	Sow summer flowers (zinnia, portulaca)	Start composting heaps
March	Sow zaid crops; harvest peas	Sow peas, spinach, potato	Harvest; summer greens	Last winter blooms	Mulch for summer
April	Water and mulch; okra, gourds	Sow beans, carrots	Shade nets for heat	Summer flowers	Water early morning
May	Heat: shade, mulch, minimal sowing	Main sowing season	Prepare for monsoon	Jasmine season	Rainwater set-up
June	Sow kharif crops with first rains	Monsoon care	Sow with rains	Plant monsoon flowers	Clear drains
July	Plant fruit trees; brinjal, chilli	Watch for fungus	Plant trees	Balsam, cockscomb	Snail watch
August	Sow carrot and radish (late August)	Harvest beans	Sow winter vegetables in hills of south	Monsoon blooms	Stake plants
September	Sow cauliflower, cabbage nurseries	Sow winter greens	Main sowing	Sow winter flower seeds	Prepare compost
October	Main rabi sowing: peas, spinach, methi, coriander	Plant bulbs	Sow	Plant winter flowers, roses	Divide perennials
November	Sow garlic, onion; transplant cauliflower	Mulch; prepare for snow	Harvest	Chrysanthemum show season	Plant roses
December	Harvest greens; protect from frost	Rest	Harvest	Flower shows	Plan next year

GRD-26 Tools, budgets and gardening with children

☐ **GRD-26.1 Basic tools**: trowel, hand fork (*khurpi*), secateurs, watering can with a rose, hose, gloves, a spade or *phawda* for beds; clean and dry tools after use.

☐ **GRD-26.2 Budget gardening**: reuse buckets, tins and cartons as pots; make compost from kitchen waste; save seeds; swap cuttings with neighbours; buy from government nurseries and KVKs.

☐ **GRD-26.3 Gardening with children**: fast crops (radish, microgreens, sunflower), a bean-in-a-jar experiment (GRD-00.02), worm bins, measuring and drawing plants; avoid toxic plants (oleander, castor, datura, gloriosa).

☐ **GRD-26.4 Safety**: tetanus protection, gloves for soil and manure, lifting pots properly, care with sprays, and sun protection.

☐ **GRD-26.5 Learning more**: KVK training, state horticulture departments, the RHS website, ICAR's guides, gardening clubs and seed festivals, and the Curious Sparoows garden diary pages.

☐ **GRD-26.6 Careers**: horticulturist, landscape architect, nursery owner, plant pathologist, botanist, agricultural officer, urban farmer; degrees from state agricultural universities and the Indian Agricultural Research Institute (Pusa).

PART 13 — REVIEWS AND VOICES

About this Part. Reviews in two forms: what experts said (summarised in our own words, with credit) and Kritikesh's own reviews of gardens, books, products and seeds.

GRD-27 Critics' Corner: great gardening books and voices

☐ **GRD-27.1 Classics**: Gilbert White, *The Natural History of Selborne* (1789); Gertrude Jekyll, *Colour in the Flower Garden* (1908); Albert Howard, *An Agricultural Testament* (1940); Rachel Carson, *Silent Spring* (1962); Fukuoka, *The One-Straw Revolution* (1975).

☐ **GRD-27.2 Modern books**: Michael Pollan, *Second Nature* (1991) and *The Botany of Desire* (2001); Robin Wall Kimmerer, *Braiding Sweetgrass* (2013); Sue Stuart-Smith, *The Well-Gardened Mind* (2019); Monty Don's books; Charles Dowding on no-dig.

☐ **GRD-27.3 India**: M. S. Randhawa, *Gardens Through the Ages* (1971) and *Flowering Trees*; Pippa Mukherjee, *Common Trees of India* (1983); the *Bombay Natural History Society* field guides.

☐ **GRD-27.4 Media**: BBC *Gardeners' World* (since 1968); RHS *The Garden* magazine; Indian gardening YouTube and Instagram communities (choose those that cite sources).

GRD-28 The experts said, I say (page template)

1. **The garden, book or product in two lines**: what, where, who designed or wrote it, when.
2. **What the experts said**: 3 to 5 named views, summarised in my own words, with links and dates.
3. **Where they agree and disagree**.
4. **What I say**: my own view (link to GRD-29).
5. **If you liked this, try next**: three linked gardens, plants or books.

GRD-29 My Review Desk (Kritikesh's garden, book and seed review template)

☐ **Header**: name, type (garden visit, book, seed variety, tool, product), place or publisher, date.

☐ **My one-line verdict**.

☐ **For gardens**: design idea, plants, upkeep, accessibility, best season, entry fee and timings (dated).

☐ **For seeds and plants**: germination rate, days to harvest, taste, pest resistance, in my conditions.

☐ **For books and tools**: who it is for, accuracy, value for money.

☐ **What I learned** and **what I would change**.

☐ **My rating** on a clear five-point scale, and **tags and try-next links**.

PART 14 — NAVIGATION: TAGS, LINKS AND THE SITE MAP

GRD-30 Tag dictionary

☐ **Kind**: #kitchengarden #fruit #herb #medicinal #flower #indoor #balcony #terrace #container #vertical #hydroponics #aquaponics #organic #naturalfarming #permaculture #xeriscape #raingarden #bonsai #succulents #watergarden #terrarium #kokedama #lawn #pollinator #therapy #schoolgarden.

☐ **Tradition**: #mughal #rajput #sacredgrove #kerala #persian #japanese #chinese #islamic #italian #french #english #cottage #dutch #moorish #african #latinamerican #naturalistic.

☐ **Plant**: one tag per plant (#tomato #tulsi #curryleaf #rose #moringa), and per family (#solanaceae #legumes).

☐ **Season**: #kharif #rabi #zaid #spring #summer #monsoon #autumn #winter.

☐ **Condition**: #soil #compost #mulch #water #drip #sun #shade #climate.

☐ **Technique**: #sowing #cuttings #layering #grafting #budding #pruning #companionplanting #croprotation #seedsaving.

☐ **Health**: #pests #diseases #ipm #deficiency.

☐ **Region**: #india and each state, and each country.

☐ **Theme**: #history #people #awards #philosophy #psychology #climate #biodiversity #foodsecurity #children.

☐ **Level**: #beginner #intermediate #expert.

GRD-31 Cross-links and the website site map

☐ **GRD-31.1 Cross-topic links**: Cooking (kitchen-garden recipes, spices: COK); Technology (smart irrigation, hydroponic sensors: TEC); Books (gardening books: BOK); Movies (garden films such as *The Secret Garden*: MOV); the Curriculum Index (Botany, Geography soils and agro-climatic zones, Environmental Studies) and the Exam Guide (RPSC/RSSB questions on agriculture, crops and soils of Rajasthan).

☐ **GRD-31.2 Suggested web addresses**: /gardening/ (home, Gateway), /gardening/seed-to-civilisation/<leaf>/ , /gardening/kinds/<kind>/ , /gardening/traditions/<tradition>/ , /gardening/plants/<plant>/ , /gardening/families/<family>/ , /gardening/native/<region>/ , /gardening/seasons/<season>/ , /gardening/soil/<topic>/ , /gardening/techniques/<technique>/ , /gardening/health/<pest-or-disease>/ , /gardening/calendar/<region>/<month>/ , /gardening/history/<era>/ , /gardening/people/<name>/ , /gardening/awards/<show>/<year>/ , /gardening/ideas/<topic>/ , /gardening/guide/<topic>/ , /gardening/reviews/<item>/ .

☐ **GRD-31.3 Search phrases to own**: "balcony vegetable garden India", "how to grow tomatoes in pots", "kitchen garden plants list", "kharif rabi zaid crops", "how to make compost at home", "yellow leaves on plants cause", "Mughal garden char bagh", "plants toxic to pets", "Chelsea Flower Show 2026 winner", with Hindi lines ("gamle mein sabzi kaise ugaye", "ghar par khad kaise banaye", "tulsi ka paudha kaise lagaye").

PART 15 — SOURCES, CREDITS AND EDITOR'S NOTES

GRD-S Sources and credits

Knowledge belongs to the people who made it. These are the sources this index stands on; every page on the website will cite its own.

☐ **Horticultural and botanical bodies**: Royal Horticultural Society (rhs.org.uk); Royal Botanic Gardens, Kew, and *Plants of the World Online* (powo.science.kew.org); Botanic Gardens Conservation International; Botanical Survey of India (bsi.gov.in); USDA (hardiness zones); FAO (soils, home gardens, IPM).

☐ **India**: ICAR and its institutes (IARI Pusa, IIHR Bengaluru, IIVR Varanasi); state agricultural universities and Krishi Vigyan Kendras; National Horticulture Board; National Medicinal Plants Board; Soil Health Card portal; Directorate of Plant Protection; Ministry of Agriculture (natural farming mission); Rashtrapati Bhavan (Amrit Udyan).

☐ **History and reference**: Encyclopaedia Britannica; M. S. Randhawa, *Gardens Through the Ages* (1971); Penelope Hobhouse, *The Story of Gardening* (2002); Stephanie Dalley, *The Mystery of the Hanging Garden of Babylon* (2013); Linnaeus (1753); Darwin (1880, 1881); UNESCO World Heritage listings.

☐ **Health and ideas**: Soga, Gaston and Yamaura, *Preventive Medicine Reports* (2017); Van den Berg and Custers, *Journal of Health Psychology* (2011); Ulrich, *Science* (1984); Kaplan and Kaplan (1989); Wilson, *Biophilia* (1984); Litt and colleagues, *Lancet Planetary Health* (2023); Cummings and Waring, *Journal of Exposure Science and Environmental Epidemiology* (2019); Lowry and colleagues, *Neuroscience* (2007).

☐ **Latest results checked (3 November 2026)**: RHS Chelsea 2026 Garden of the Year (Sarah Eberle, CPRE "On the Edge") and Plant of the Year (*Hosta* 'Red Ninja'), from *Gardens Illustrated* and rhs.org.uk; press reports on Saalumarada Thimmakka (November 2025).

GRD-E Editor's notes (corrections and decisions on the brief)

1. **The Hanging Gardens of Babylon** are presented as unlocated, with Dalley's Nineveh theory noted.
2. **"Mughal gardens" were renamed Amrit Udyan** in 2023; both names are used so readers can search either.
3. **Mehrgarh is in Pakistan's Balochistan**; it is part of South Asia's shared heritage.
4. **Indoor plants and clean air**: the NASA claim is corrected with later evidence (GRD-04.1d).
5. **Natural farming, biodynamics and traditional ferments** are described fairly, with a note where scientific evidence is limited or debated.
6. **Medicinal plants** carry a caution: growing them is gardening; using them as medicine needs qualified advice.
7. **The 13-field plant card** is split into two linked tables (identity and care) so it fits a printed page.
8. **Added beyond the brief**: mushrooms and microgreens, protected cultivation, nutrient-symptom logic, invasive species, seed rights law, toxic plants for children and pets, careers, and indigenous knowledge.
9. **Not stated because not verified**: earlier Chelsea Best in Show winners year by year; they will be added from the RHS archive on the website.

GRD-T My Gardening Tracker

Use the ☐ boxes across the booklet to tick what you have grown, visited or learned. A simple tracker for each Part: Part · Sections · Explored · Date started · Date finished. Keep a separate garden diary: date, crop, variety, sown, sprouted, first harvest, notes.

Progress tracker

Part	Sections	Explored ☐	Started	Finished
PART 0 — THE INTRODUCTION TWIG: HOW DID A SEED IN THE	8			

SOIL BECOME THE FIRST STEP OF EVERY CIVILISATION?				
PART 1 — THE MASTER MAP OF GARDENING	1			
PART 2 — KINDS OF GARDENS	3			
PART 3 — GARDEN TRADITIONS OF INDIA AND THE WORLD	2			
PART 4 — THE PLANT WORLD FOR GARDENERS	3			
PART 5 — SOIL, WATER, LIGHT AND CLIMATE	3			
PART 6 — TECHNIQUES	2			
PART 7 — PLANT HEALTH	2			
PART 8 — REGISTERS	2			
PART 9 — HISTORY	1			
PART 10 — PEOPLE, SHOWS AND AWARDS	2			
PART 11 — IDEAS: PHILOSOPHY, PSYCHOLOGY AND SOCIETY	2			
PART 12 — THE PRACTICAL GUIDE	3			
PART 13 — REVIEWS AND VOICES	3			
PART 14 — NAVIGATION: TAGS, LINKS AND THE SITE MAP	2			
PART 15 — SOURCES, CREDITS AND EDITOR'S NOTES	3			